

LNS: Approximation Methods and Extended Precision Formats

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Abstract

This document extends previous work on the Logarithmic Number System (LNS) arithmetic unit for RISC-V, which demonstrated a 16-bit LNS format achieving IEEE-754 equivalent precision for neural network weights using linear spline approximation with a greedy algorithm, requiring only 324 bytes of memory and synthesizing at 125 MHz on FPGA.

The present work broadens this foundation in two directions. First, it revisits lns8 Q4.3 and lns16 Q8.7 with a systematic comparison of approximation methods beyond linear splines, including Newton's divided differences, minimax polynomial approximation via the Remez algorithm, and piecewise Chebyshev approximation. Second, it extends the LNS framework to lns32 Q16.15 and lns64 Q32.31, where the domain of the correction functions $f^+(x) = \log_2(1 + 2^x)$ and $f^-(x) = \log_2(1 - 2^x)$ grows significantly and spline table sizes become impractical, motivating the transition to compact polynomial approximations.

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List of Acronyms

NN	Neural Network
FP	Floating Point
LNS	Logarithmic Number System
LNSU	Logarithmic Number System Unit
LUT	Look Up Table
DSP	Digital Signal Processing block/unit
XF	Spline format storing (x, f) point pairs
XMB	Spline format storing (x, m, b) segment coefficients
NDD	Newton's Divided Differences
MAE	Maximum Absolute Error
MRE	Maximum Relative Error
RMSE	Root Mean Square Error

1 Introduction

The Logarithmic Number System (LNS) offers a compelling alternative to IEEE-754 floating point. By encoding values as fixed-point binary logarithms, multiplication and division reduce to integer addition and subtraction, and square root reduces to a single arithmetic right shift. These properties eliminate the need for mantissa multipliers in hardware, significantly reducing area and power consumption.

Previous work demonstrated an lns16 Q8.7 arithmetic unit embedded in a RISC-V core, achieving a maximum one-bit approximation error for addition and subtraction using linear splines generated by a greedy algorithm, with a total memory footprint of 324 bytes. The design synthesized at 125 MHz on a Pynq Z2 FPGA with single-cycle latency.

This document builds on that foundation. The first goal is to evaluate alternative approximation methods for the correction functions required by LNS addition and subtraction on lns8 and lns16 formats, characterising the trade-off between approximation accuracy, evaluation cost, and memory footprint. The second goal is to extend the LNS framework to 32-bit and 64-bit formats, where the growing domain of the correction functions makes the linear spline approach impractical and more compact polynomial representations become necessary.

1.1 Scope

This document covers:

- A review of the LNS format and the derivation of the correction functions for add and sub operations.
- A summary of the previous linear spline (XF and XMB) approach and its results for lns8 and lns16.
- An analysis of the correction functions' behaviour at increased precision and wider domains (lns32, lns64).
- A survey and analysis of candidate approximation methods: Newton's divided differences, minimax polynomial approximation and piecewise Chebyshev approximation.
- A comparison framework for evaluating approximation methods across all four formats.
- Open questions and planned implementation work.

2 Background

2.1 The Logarithmic Number System

In LNS, a non-zero real number x is represented by a sign bit S_x and a fixed-point binary logarithm L_x :

$$\begin{aligned} S_x &= \begin{cases} 1, & \text{if } x < 0 \\ 0, & \text{if } x \geq 0 \end{cases} \\ L_x &= \log_2(|x|) \\ x &= (-1)^{S_x} \cdot 2^{L_x} \end{aligned} \tag{1}$$

L_x is a two's complement fixed-point number. Negative values of L_x represent magnitudes in $(0, 1)$ and positive values represent magnitudes in $(1, +\infty)$.

2.2 Supported Formats

The previous supported LNS formats use one sign bit and a fixed-point exponent split into integer and fractional parts:

Format	S_x	Integer bits	Fractional bits	Value range
lns8 Q4.3	1	4	3	2^{-8} to $2^{7.875}$
lns16 Q8.7	1	8	7	2^{-128} to $2^{127.992}$

Table 1: LNS format specifications

The fractional resolution per step is 2^{-F} where F is the number of fractional bits, giving a relative quantization error of $2^{2^{-F}} - 1$. For lns8 this is $2^{-3} = 0.125$, with a relative quantization error per step of approximately $2^{\frac{1}{8}} - 1 \approx 9\%$. For lns16 it is $2^{-7} = 0.0078$, approximately $2^{\frac{1}{128}} - 1 \approx 0.54\%$.

2.3 Arithmetic Operations

The operations in LNS follow directly from the logarithmic representation:

$$\begin{aligned} z = x \cdot y &\Leftrightarrow L_z = L_x + L_y \\ z = \frac{x}{y} &\Leftrightarrow L_z = L_x - L_y \\ z = \sqrt{x} &\Leftrightarrow L_z = L_x \gg 1 \\ z = x + y &\Leftrightarrow L_z = L_x + \log_2(1 + 2^{L_y - L_x}), \quad |x| \geq |y| \\ z = x - y &\Leftrightarrow L_z = L_x + \log_2(1 - 2^{L_y - L_x}), \quad |x| \geq |y| \end{aligned} \tag{2}$$

Multiplication, division, and square root are exact up to quantization. Addition and subtraction require evaluating the correction functions:

$$\begin{aligned} f^+(x) &= \log_2(1 + 2^x), \quad x \in (-\infty, 0] \\ f^-(x) &= \log_2(1 - 2^x), \quad x \in (-\infty, 0) \end{aligned} \tag{3}$$

where $x = L_y - L_x \leq 0$ is enforced by swapping operands if necessary so that the larger magnitude is always taken as the base.

This comes from the following derivation:

- Given $|x| > |y|$:

$$\begin{aligned}
 z &= x \pm y \\
 L_z &= \log_2(2^{L_x} \pm 2^{L_y}) \\
 L_z &= \log_2(2^{L_x}(1 \pm 2^{L_y-L_x})) \\
 L_z &= L_x + \log_2(1 \pm 2^{L_y-L_x})
 \end{aligned} \tag{4}$$

2.4 Properties of the Correction Functions

2.4.1 Addition: $f^+(x) = \log_2(1 + 2^x)$

f^+ is smooth and bounded on $(-\infty, 0]$. It increases monotonically from 0 at $-\infty$ to 1 at $x = 0$. Its first and second derivative are:

$$\begin{aligned}
 (f^+)'(x) &= \frac{2^x}{1 + 2^x} \\
 (f^+)'(x) &= \ln(2) \frac{2^x}{(1 + 2^x)^2}
 \end{aligned} \tag{5}$$

Both are always positive, monotonically increasing, and approach 0 at $-\infty$ and $\frac{1}{2}$ and $\frac{\ln(2)}{4}$ at $x = 0$, respectively. These derivatives will be important to derive upper bound expressions for the error functions.

2.4.2 Subtraction: $f^-(x) = \log_2(1 - 2^x)$

f^- is defined on $(-\infty, 0)$ only. As $x \rightarrow 0^-$, $f^-(x) \rightarrow -\infty$ due to the singularity of $\log_2$ at zero. Its first and second derivative are:

$$\begin{aligned}
 (f^-)'(x) &= -\frac{2^x}{1 - 2^x} \\
 (f^-)'(x) &= -\ln(2) \frac{2^x}{(1 - 2^x)^2}
 \end{aligned} \tag{6}$$

These are always negative and diverge to $-\infty$ as $x \rightarrow 0^-$. The unbounded second derivative near $x = 0$ is the fundamental reason that the subtraction function requires significantly more intervals than the addition function to achieve the same approximation error. Any approximation method must handle this singularity explicitly, either by excluding a neighbourhood of 0 and clamping, or by using a change of variable that regularises the behaviour.

3 Previous Work: Linear Splines

3.1 XF and XMB Representations

The previous work approximated f^+ and f^- with piecewise linear splines generated by a greedy algorithm. Given a budget of $n + 1$ points $\{(x_i, f_i)\}_{i=0}^{\{n\}}$, the standard linear spline interpolation formula for an interval $[x_{i-1}, x_i]$ is:

$$\text{LS}_i(x) = \frac{(x_i - x)f_{i-1} + (x - x_{i-1})f_i}{h_i}, \quad h_i = x_i - x_{i-1} \quad (7)$$

This is the XF representation. It stores (x_i, f_i) pairs and evaluates the weighted average at query time. An equivalent form is:

$$\text{LS}_i(x) = m_i \cdot x + b_i, \quad m_i = \frac{f_i - f_{i-1}}{h_i}, \quad b_i = \frac{f_{i-1}x_i - f_i x_{i-1}}{h_i} \quad (8)$$

The error upper bound formula for a linear spline is:

$$\varepsilon \leq \frac{1}{8} \max_{[a,b]} |f''(x)| h_i^2 \quad (9)$$

This is the XMB representation. It precomputes the slope m_i and intercept b_i per segment, reducing runtime evaluation to a single multiply-add. The trade-off is increased memory: XF requires $4(n + 1)$ bytes for `lms16` (two 2-byte fields per point), while XMB requires $6(n + 1)$ bytes (three 2-byte fields).

3.2 Greedy Algorithm

The greedy algorithm starts with the two boundary points and iteratively adds the midpoint of the interval with the largest error estimate:

$$\varepsilon_i \leq \frac{\ln(2) \cdot 2^{x_i} \cdot h_i^2}{8(1 \pm 2^{x_i})^2} \quad (10)$$

Since the error bound is monotonically increasing for both functions (the maximum of $|f''|$ on each interval is at the right endpoint), the interval to split is always the one with the largest h_i weighted by the local curvature. The minimum allowed interval length is 2 (in fixed-point units), enforcing that consecutive spline points are distinct in the integer representation.

3.3 Results for lns8 and lns16

Format	Method	Operation	Error (bits)	Table size (bytes)
lns8 Q4.3	XF	Add	1	16
		Sub	1	48
		Add+Sub	1	64
	XMB	Add	1	24
		Sub	1	42
		Add+Sub	1	66
lns16 Q8.7	XF	Add	1	40
		Sub	1	200
		Add+Sub	1	240
	XMB	Add	1	72
		Sub	1	252
		Add+Sub	1	324

Table 2: Linear spline results for lns8 Q4.3 and lns16 Q8.7

The XF tables achieve lower memory for the same error, while XMB reduces runtime to a single multiply-add plus a table lookup, enabling single-cycle latency.

The subtraction table requires substantially more rows than the addition table, because the second derivative of f^- diverges near $x = 0$, requiring dense sampling near the singularity.

3.3.1 f^+ and f^- domain ranges

For an LNS type with precision p , the minimum value in absolute terms is 2^{-p} . So, we want to find $x < 0$, such that $\pm 2^{-p} = \log_2(1 \pm 2^x)$:

$$\begin{aligned}
\pm 2^{-p} &= \log_2(1 \pm 2^x) \\
2^{\pm 2^{-p}} &= 1 \pm 2^x \\
2^{\pm 2^{-p}} - 1 &= \pm 2^x \\
\pm(2^{\pm 2^{-p}} - 1) &= 2^x \\
x &= \log_2(\pm(2^{\pm 2^{-p}} - 1))
\end{aligned}$$

Type	Precision p (bits)	$x_{\min}^+$	$x_{\min}^-$
LNS8 Q4.3	3	-3.466	-3.591
LNS16 Q8.7	7	-7.525	-7.533
LNS32 Q16.15	15	-15.529	-15.529
LNS32 Q8.23	23	-23.529	-23.529
LNS64 Q33.30	30	-30.529	-30.529
LNS64 Q23.40	40	-40.529	-40.529
LNS64 Q11.52	52	-52.000	-53.000

Table 3: Effective domain lower bound $x_{\min}^{\pm} = \log_2(2^{\pm 2^{-p}} - 1)$ by format and precision

3.3.2 FP to LNS and LNS to FP conversion functions

$$\begin{aligned}
 x_{fp} &= (-1)^{S_x} \cdot (1 + m_x) \cdot 2^{e_x} \\
 x_{lns} &= (-1)^{S_x} \cdot 2^{L_x} \\
 x_{fp} &= x_{lns} \\
 (-1)^{S_x} \cdot (1 + m_x) \cdot 2^{e_x} &= (-1)^{S_x} \cdot 2^{L_x} \\
 (1 + m_x) \cdot 2^{e_x} &= 2^{L_x} \\
 \log_2(1 + m_x) + e_x &= L_x \\
 L_x &= e_x + \log_2(1 + m_x) \\
 \text{float2lns}(m_x) &:= \log_2(1 + m_x) \\
 \text{lns2float}(\text{lns_f}_x) &:= 2^{\text{lns_f}_x} - 1
 \end{aligned} \tag{11}$$

$$\begin{aligned}
 \frac{d^2}{dx^2} \text{float2lns}(x) &= \frac{d}{dx} \left(\frac{1}{\ln(2)} \frac{1}{1+x} \right) \\
 &= -\frac{1}{\ln(2)} \frac{1}{(1+x)^2} \\
 \frac{d^2}{dx^2} \text{lns2float}(x) &:= \ln(2)^2 2^x
 \end{aligned}$$

$$\begin{aligned}
 \varepsilon_{\text{float2lns}} &\leq \frac{1}{8} \frac{1}{\ln(2)} \frac{1}{(1+x_{i-1})^2} h_i^2 \\
 \varepsilon_{\text{lns2float}} &\leq \frac{1}{8} \ln(2)^2 2^{x_i} h_i^2
 \end{aligned} \tag{12}$$

3.4 Limitations at Higher Precision

The linear spline approach works well for `lns8` and `lns16`, because the domain of the correction functions is small — the argument $x = L_y - L_x$ ranges over at most 8 or 16 bits, respectively. For `lns32` and `lns64`, the domain grows proportionally with the exponent width, and the number of intervals required to maintain a fixed error bound scales roughly as $O\left(\sqrt{\frac{1}{\varepsilon}}\right) = O\left(\sqrt{\frac{1}{2^{-n}}}\right) = O(2^{\frac{n}{2}})$ per unit of domain, making table-based approaches impractical.

Additionally, even for `lns8` and `lns16` there is room to improve approximation quality beyond piecewise linear. Higher-degree polynomial approximations can achieve the same error with fewer segments or lower error with the same number of segments.

4 Candidate Approximation Methods

This chapter surveys the methods under consideration for replacing or supplementing linear splines for `lms8`, `lms16`, and for use in `lms32` and `lms64`.

4.1 Error Formula

Theorem 1 Lagrange Interpolation Error Formula

Given a function $f(x) \in C^{n+1}[a, b]$, let $p_n(x) \in \mathbb{P}_n$ be the interpolative polynomial of $f(x)$ in the unique x 's $\{x_i \mid 0 \leq i \leq n\}$, then $\forall x \in [a, b], \exists c_x \in [a, b]$:

$$\varepsilon = f(x) - p_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c_x) \pi_{n+1}(x) \quad (13)$$

where $\pi_{n+1}(x) = \prod_{k=0}^n (x - x_k)$

4.1.1 f^+ and f^- Error Upper Bound

Given $x < 0$:

$$\begin{aligned} \log_2(1 \pm 2^x) &= \frac{1}{\ln(2)} \ln(1 \pm 2^x) \\ &= \frac{1}{\ln(2)} \int_{-\infty}^x \frac{d}{dt} (\ln(1 \pm 2^t)) dt \\ &= \frac{1}{\ln(2)} \int_{-\infty}^x \frac{1}{1 \pm 2^t} \frac{d}{dt} (1 \pm 2^t) dt \\ &= \frac{\pm}{\ln(2)} \int_{-\infty}^x \frac{2^t}{1 \pm 2^t} dt \end{aligned}$$

As $\pm 2^t < 0, \forall t \in (-\infty, x], x < 0$, then $\frac{1}{1 \pm 2^t}$ can be expanded using the geometric series infinite series representation $\frac{1}{1-r} = \sum_{i \geq 0} r^i$.

$$= \frac{\pm}{\ln(2)} \int_{-\infty}^x 2^t \sum_{n \geq 0} (\mp 2^t)^n dt$$

Since the series converges, the swap of the integration and summation operations are justified.

$$\begin{aligned} &= \frac{\pm}{\ln(2)} \sum_{n \geq 0} (\mp 1)^n \int_{-\infty}^x 2^{(n+1)t} dt \\ &= \frac{1}{\ln(2)} \sum_{n \geq 0} (\mp 1)^{n-1} \left(\frac{1}{\ln(2)(n+1)} 2^{(n+1)t} \right) \Bigg|_{t \rightarrow -\infty}^{t=x} \\ &= \frac{1}{\ln(2)^2} \sum_{n \geq 0} \frac{(\mp 1)^{n-1}}{n+1} 2^{(n+1)x} \\ &= \frac{1}{\ln(2)^2} \sum_{n \geq 1} \frac{(\mp 1)^n}{n} 2^{nx} \end{aligned}$$

And so, we finally get an infinite series representation of f^+ and f^- .

$$\log_2(1 \pm 2^x) = \frac{1}{\ln(2)^2} \sum_{n \geq 1} \frac{(\mp 1)^n}{n} 2^{nx}, x < 0 \quad (14)$$

Now, we can use this infinite series representation to calculate an upper bound for the k -th derivative of both functions:

$$\frac{d^k}{dx^k} \log_2(1 \pm 2^x) = \frac{1}{\ln(2)^2} \frac{d^k}{dx^k} \sum_{n \geq 1} \frac{(\mp 1)^n}{n} 2^{nx}$$

The switch of the summation and derivative operators is also justified as the series does converge.

$$\begin{aligned} \frac{d^k}{dx^k} \log_2(1 \pm 2^x) &= \frac{1}{\ln(2)^2} \sum_{n \geq 1} \frac{(\mp 1)^n}{n} \frac{d^k}{dx^k} 2^{nx} \\ \frac{d^k}{dx^k} \log_2(1 \pm 2^x) &= \ln(2)^{k-2} \sum_{n \geq 1} (\mp 1)^n n^{k-1} 2^{nx} \end{aligned}$$

If we can find a term a , such that all terms $n^{k-1} 2^{nx} \leq a^n$, then the sum of the terms will be larger than our current series and we have a simple upper bound we can compute.

Following the Alternative Series Estimation Error for f^+ ,
as $|a_{n+1}| < |a_n| \wedge \lim_{n \rightarrow \infty} a_n = 0, a_n = (-1)^n n^{k-1} 2^{nx}$:

$$\begin{aligned} |a_{n+1}| &< |a_n| \\ |(n+1)^{k-1} 2^{(n+1)x}| &< |n^{k-1} 2^{nx}| \\ \left(1 + \frac{1}{n}\right)^{k-1} &< 2^{-x} \\ 1 + \frac{1}{n} &< 2^{-\frac{x}{k-1}} \\ \frac{1}{n} &< 2^{-\frac{x}{k-1}} - 1 \\ n &> \frac{1}{2^{-\frac{x}{k-1}} - 1} \end{aligned}$$

$$\begin{aligned}
 |R_n| &< |a_{n+1}| \\
 \left| \sum_{n \geq \left\lceil \frac{1}{2^{-\frac{x}{k-1}} - 1} \right\rceil} (-1)^n n^{k-1} 2^{nx} \right| &< \left| \left(\left\lceil \frac{1}{2^{-\frac{x}{k-1}} - 1} \right\rceil + 1 \right)^{k-1} 2^{(1+1)x} \right| \\
 &= \left(\left\lceil \frac{1}{2^{-\frac{x}{k-1}} - 1} \right\rceil + 1 \right)^{k-1} 2^{2x} \\
 &= \left(\left\lceil \frac{2^{\frac{x}{k-1}}}{1 - 2^{\frac{x}{k-1}}} \right\rceil + 1 \right)^{k-1} 2^{2x} \\
 \left\lceil \frac{2^t}{1 - 2^t} \right\rceil + 1 &\leq \frac{2^t}{1 - 2^t} + 2 \\
 &= \frac{2^t + 2 - 2 \cdot 2^t}{1 - 2^t} \\
 &= \frac{2 - 2^t}{1 - 2^t} \\
 &= 2 \frac{1 - 2^{t-1}}{1 - 2^t} \\
 x < 0 &\rightarrow \frac{x}{k-1} < 0 \rightarrow t < 0 \\
 2 \frac{1 - 2^{t-1}}{1 - 2^t} &\leq \frac{1}{1 - 2^t} \\
 \left(\frac{1}{1 - 2^{\frac{x}{k-1}}} \right)^{k-1} &= \sum_{n \geq 0} 2^{\frac{nx}{k-1}} \\
 K &= \left| \sum_{n \geq 1}^{n = \left\lceil \frac{1}{1 - 2^{\frac{x}{k-1}}} \right\rceil - 1} (-1)^n n^{k-1} 2^{nx} \right|
 \end{aligned}$$

Therefore, the k-th derivative of the f^+ is bounded by:

$$\frac{d^k}{dx^k} \log_2(1 + 2^x) \leq \ln(2)^{k-2} \left(K + \left(1 - 2^{\frac{x}{k-1}}\right)^{1-k} 2^{2x} \right)$$

Following D'Alambert Criteria for f^- , as $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$, $a_n = \frac{2^{nx}}{n^{k+1}}$:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{(n+1)^{k-1} 2^{(n+1)x}}{n^{k-1} 2^{nx}} &= \lim_{n \rightarrow \infty} 2^{(n+1)x-nx} \left(\frac{n+1}{n} \right)^{k+1} \\ &= 2^x \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{k+1} \\ &= 2^x \end{aligned}$$

$$\begin{aligned} L < 1 &\Leftrightarrow 2^x < 1 \\ &\Leftrightarrow x < 0 \end{aligned}$$

$$\begin{aligned} R_n &\leq a_{n+1} \frac{1}{1-L} \\ \sum_{n \geq 1} \frac{2^{nx}}{n^{k+1}} &\leq 2^{k-1} 2^{2x} \frac{1}{1-2^x} \\ &= \frac{2^{2x+k-1}}{1-2^x} \end{aligned}$$

Therefore, the k -th derivative of the f^- is bounded by:

$$\begin{aligned} \frac{d^k}{dx^k} \log_2(1-2^x) &\leq \ln(2)^{k-2} \frac{2^{2x+k-1}}{1-2^x} \leq \ln(2)^{k-2} \left(K + \left(1 - 2^{\frac{x}{k-1}} \right)^{1-k} 2^{2x} \right) \\ \frac{2^{2x+k-1}}{1-2^x} &\leq \left(1 - 2^{\frac{x}{k-1}} \right)^{1-k} 2^{2x} \\ &\Leftrightarrow \frac{2}{1-2^x} \leq \frac{1}{1-2^{\frac{x}{k-1}}} \\ &\Leftrightarrow 2 - 2 \cdot 2^{\frac{x}{k-1}} \leq 1 - 2^x \\ &\Leftrightarrow 2 \cdot 2^{\frac{x}{k-1}} - 2^x \geq 1 \\ &\Leftrightarrow 2^x \left(2^{x(\frac{1}{k-1}-1)} - 1 \right) \geq 2^{-1} \end{aligned}$$

Now we can use the Lagrange Interpolation Error Formula with $n+1$ points.

$$\begin{aligned} \varepsilon^+ &\leq \max_{x \in [a,b]} \frac{1}{(n+1)!} \ln(2)^{(n+1)-2} \left(K + \left(1 - 2^{\frac{x}{(n+1)-1}} \right)^{1-(n+1)} 2^{2x} \right) |\pi_{n+1}(x)| \\ \varepsilon^- &\leq \max_{x \in [a,b]} \frac{1}{(n+1)!} \ln(2)^{(n+1)-2} \frac{2^{2x+(n+1)-1}}{1-2^x} |\pi_{n+1}(x)| \end{aligned}$$

For an interval $[a, b]$, $\{x_i \mid 0 \leq i \leq n\} \in [a, b]$, where n is the degree of the polynomial on the interval $[a, b]$ and $x \in [a, b]$ and $x < 0$:

$$\begin{aligned} \varepsilon_n^+ &\leq \max_{x \in [a,b]} \frac{1}{(n+1)!} \ln(2)^{n-1} \left(K + \left(1 - 2^{\frac{x}{n}} \right)^{-n} 2^{2x} \right) (b-a)^{n+1} \\ \varepsilon_n^- &\leq \max_{x \in [a,b]} \frac{1}{(n+1)!} \ln(2)^{n-1} \frac{2^{2x+n}}{1-2^x} (b-a)^{n+1} \\ \varepsilon_n^+ &\leq \frac{1}{(n+1)!} \ln(2)^{n-1} \left(K + \left(1 - 2^{\frac{x}{n}} \right)^{-n} 2^{2x} \right) (b-a)^{n+1} \\ \varepsilon_n^- &\leq \frac{1}{(n+1)!} \ln(2)^{n-1} \frac{2^{2b+n}}{1-2^b} (b-a)^{n+1} \end{aligned}$$

Splitting the functions into multiple branches with intervals $\{[x_{i-1}, x_i] \mid 0 \leq i \leq k\}$ with degree n_i :

$$\begin{aligned}\varepsilon_{n_i}^+ &\leq \frac{1}{(n_i + 1)!} \ln(2)^{n_i-1} \left(K + \left(1 - 2^{\frac{x}{n_i}} \right)^{-n_i} 2^{2x} \right) h_i^{n_i+1} \\ \varepsilon_{n_i}^- &\leq \frac{1}{(n_i + 1)!} \ln(2)^{n_i-1} \frac{2^{2x_i+n_i}}{1 - 2^{x_i}} h_i^{n_i+1}\end{aligned}\tag{15}$$

Now, we have an upper bound that we can use to modulate the approximation error of both functions in all k sections.

But we can go further. Let p be the precision of our LNS format, that is, LNS n -Qi. p , we want $\varepsilon_{n_i} \leq 2^{-p}$:

$$\begin{aligned}2^{-p^+} &\leq \frac{1}{(n_i + 1)!} \ln(2)^{n_i-1} h_i^{n_i+1} \left(K + \left(1 - 2^{\frac{x}{n_i}} \right)^{-n_i} 2^{2x} \right) \\ p^+ &\geq \log_2 \left((n_i + 1)! \cdot \ln(2)^{1-n_i} \cdot h_i^{-(n_i+1)} \cdot 2^{-2x_i-n_i} \right) \\ p^+ &\geq \log_2((n_i + 1)!) + (1 - n_i) \log_2(\ln(2)) - (n_i + 1) \log_2(h_i) + \log_2(2^{-2x_i-n_i}) \\ p^+ &\geq \log_2((n_i + 1)!) + (n_i + 1)(\log_2(h_i^{-1}) - \log_2(\ln(2)) - 1) - 2x_i + 2 \log_2(\ln(2)) \\ 2^{-p^-} &\leq \frac{1}{(n_i + 1)!} \ln(2)^{n_i-1} h_i^{n_i+1} 2^{2x_i+n_i} \frac{1}{1 - 2^{x_i}} \\ p^- &\geq \log_2 \left((n_i + 1)! \cdot \ln(2)^{1-n_i} \cdot h_i^{-(n_i+1)} \cdot 2^{-2x_i-n_i} \cdot (1 - 2^{x_i}) \right) \\ p^- &\geq \log_2((n_i + 1)!) + (1 - n_i) \log_2(\ln(2)) - (n_i + 1) \log_2(h_i) + \log_2(2^{-2x_i-n_i}) + \log_2(1 - 2^{x_i}) \\ p^- &\geq \log_2((n_i + 1)!) + (n_i + 1)(\log_2(h_i^{-1}) - \log_2(\ln(2)) - 1) - 2x_i + \log_2(1 - 2^{x_i}) + 2 \log_2(\ln(2))\end{aligned}$$

Finally, we arrive at a relationship between the precision of the format, the degree n_i of the polynomial, the size of the interval, h_i , and, given by the last term, the curvature of the function in that same interval.

$$p^+ \sim \Omega(\log_2((n_i + 1)!) + (n_i + 1) \log_2(h_i^{-1}) - 2x_i)\tag{16}$$

$$p^- \sim \Omega(\log_2((n_i + 1)!) + (n_i + 1) \log_2(h_i^{-1}) - 2x_i + \log_2(1 - 2^{x_i}))\tag{17}$$

4.1.2 LNS-Float & Float-LNS Conversion Functions Aproximations

$$\text{float2lns}(x) := \log_2(1+x) = \frac{1}{\ln(2)} \sum_{n \geq 1} \frac{(-1)^n}{n} x^n, |x| < 1$$

$$\frac{d^k}{dx^k} \log_2(1+x) = \frac{1}{\ln(2)} \frac{d^k}{dx^k} \sum_{n \geq 1} \frac{(-1)^n}{n} x^n$$

$$\frac{d^k}{dx^k} \log_2(1+x) = \frac{1}{\ln(2)} \sum_{n \geq 1} \frac{(-1)^n}{n} \frac{n!}{(n-k)!} x^{n-k}$$

$$\left| \frac{(-1)^{n+1}}{n+1} \frac{(n+1)!}{(n+1-k)!} x^{n+1-k} \right| < \left| \frac{(-1)^n}{n} \frac{n!}{(n-k)!} x^{n-k} \right|$$

$$\frac{1}{n+1} \frac{(n+1)!}{(n+1-k)!} x^{n+1-k} < \frac{1}{n} \frac{n!}{(n-k)!} x^{n+1-k}$$

$$nx < (n+1-k)$$

$$x < \frac{n+1-k}{n}$$

$$x < 1 - \frac{k-1}{n}$$

$$n > \frac{k-1}{1-x}$$

$$\left| \frac{d^k}{dx^k} \log_2(1+x) \right| \leq \left| \frac{1}{\ln(2)} \sum_{n=1}^{n=i} \frac{(-1)^n}{n} \frac{n!}{(n-k)!} x^{n-k} \right| + \left| \frac{1}{\ln(2)} \frac{(-1)^{i+1}}{i+1} x^{i+1} \right|, i = \left\lceil \frac{k-1}{1-x} \right\rceil$$

By the Alternating Series Test.

$$\varepsilon_i \leq \frac{1}{(n_i+1)!} \frac{1}{\ln(2)} \frac{1}{n_i+2} x^{n_i+1} h_i^{n_i+1}$$

$$p \geq \Omega(\log_2((n_i+1)!) + (n_i+1)(\log_2(h_i^{-1}) - \log_2(x_i)) - x_i)$$

$$\text{lns2float}(x) := 2^x - 1, |x| < 1$$

$$\left| \frac{d^k}{dx^k} (2^x - 1) \right| = \ln(2)^k 2^x$$

$$\varepsilon_i \leq \frac{1}{(n_i+1)!} \ln(2)^{n_i+1} 2^{x_i} h_i^{n_i+1}$$

$$p \geq \Omega(\log_2((n_i+1)!) + (n_i+1)(\log_2(h_i^{-1}) - \log_2(\ln(2))) - x_i)$$

4.2 Newton's Divided Differences

4.2.1 Method

Newton's divided differences construct an interpolating polynomial of degree n through $n + 1$ sample points $\{(x_i, f_i) \mid 0 \leq i \leq n\}$. The polynomial is expressed in Newton form:

$$P(x) = \sum_{i=0}^n [y_0, \dots, y_i] \prod_{j=0}^{i-1} (x - x_j) \quad (18)$$

where $[y_0, \dots, y_k]$ denotes the k -th order divided difference, computed recursively as:

$$\begin{aligned} [y_i] &= f(x_i) \\ [y_i, \dots, y_{i+k}] &= \frac{[y_{i+1}, \dots, y_{i+k}] - [y_i, \dots, y_{i+k-1}]}{x_{i+k} - x_i} \end{aligned} \quad (19)$$

Evaluation via Horner's method requires n multiplications and n additions for a degree- n polynomial, which is optimal.

4.2.2 Application to LNS

For f^+ and f^- , sample points can be placed non-uniformly, with higher density near $x = 0$ where the functions curve sharply, and sparser sampling in the flat tail near $-\infty$. This is the key advantage over uniform splines: the same number of points can cover a much wider domain with controlled error.

For a piecewise approach, the domain is split into segments and a separate low-degree Newton polynomial is fitted to each. Segment boundaries are chosen adaptively based on the local curvature.

4.2.3 Memory and Evaluation Cost

Storing $n + 1$ points requires $2(n + 1)$ values: the x_i ones and the divided difference coefficients. For a degree- d polynomial on k segments, the total storage is $O(k \cdot d)$ values. Evaluation requires d multiply-adds per query.

4.2.4 Considerations

Newton's divided differences are numerically stable when using Horner's scheme, but the divided differences themselves can suffer from cancellation when computed in floating point for closely spaced points. For fixed-point arithmetic, careful scaling of the coefficients is needed to avoid overflow during evaluation.